

NANCY JIA CHEN

✉ nancy.chen@tufts.edu  [linkedin.com/in/nancy-j-chen](https://www.linkedin.com/in/nancy-j-chen)  nancy-chen-portfolio.netlify.app/

Education

Tufts University

Medford, MA

B.S. Computer Science and Mathematics, with concentration in Computer Engineering

Sep. 2023 – May 2027

GPA: 3.7, Dean's List

Relevant Courses: Algorithms, Reinforcement Learning, Operating Systems, Software Performance Engineering

Hong Kong University of Science and Technology (HKUST)

Kowloon, Hong Kong

Exchange student in the School of Engineering

Jan 2026 – May 2026

Skills

Languages: Typescript, JavaScript, Python, Java, C, C++, Assembly, Bash, HTML/CSS, SQL, Linux, Perl, SystemVerilog

Frameworks: Next.js, React.js, Bootstrap, NumPy, Pandas, Scikit-Learn, Java Swing, Robot Operating System (ROS)

Dev Tools: VSCode, Docker, Flask, Wireshark, Solidworks, AutoCAD, IntelliJ, Git, Figma, Trello, ChatGPT, Claude

Experience

Tufts University StAAR Center

Medford, MA

Lead Subject Tutor

Aug 2024 – Present

- Facilitate training for new tutors by providing Tutorfinder portal logistics, practicing roleplay scenarios to prep for future tutoring sessions, and sharing advice based on personal experience.
- Provide one-on-one and drop-in tutoring to undergraduate students in CS and math, adapting communication style to meet diverse learning needs and help students build independent academic strategies.

Break Through Tech

Boston, MA

AI Scholar

Jun 2024 – Dec 2024

- Built and optimized machine learning models (Decision Trees, Logistic Regression, Random Forest) using Python (NumPy, Pandas, Scikit-Learn), enhancing expertise in bias-variance trade-offs and model evaluation
- Developed a Natural Language Processing model to analyze Airbnb host descriptions and predict host acceptance rates.

Amazon Advertising

Manhattan, NY

Data Analyst Apprentice

Oct 2021 – Oct 2024

- Streamlined content and removed outdated pages on the Insights Team's internal website using HTML, CSS, and XWiki
- Enhanced SQL queries for ad product insights from proprietary databases, improving brand awareness rates.
- Aggregated and analyzed 1000+ rows of audience data from Tableau and Kantar market research platforms by creating cross-tabulations in Excel to identify consumer demographics, interests, and brand interaction patterns.

Research

Summer Global Research in Taiwan Program

Tainan, Taiwan

Student Researcher at Department of Biotechnology and Bioindustry Services

Jun 2026 – Aug 2026

- Conducted research on an gene alignment pipeline under the supervision of Professor Tsung-Lin Liu.
- Quality checked pipeline codebase containing Perl and C programming languages by ensuring program executions matches that of biological functions within the immune system and peer-reviewing my findings with my lab mentors.
- Implemented, tested, and documented solutions for three pre-existing issues in the pipeline codebase by reviewing my modifications and uncertainties with my PI during our weekly progress meetings.

Research Experiences for Undergraduates (REU) Program in Data Science

Medford, MA

Student Researcher at Multimodal Learning, Interaction, and Perception Lab

Jun 2024 – Aug 2024

- Conducted NSF-sponsored research on Augmented Reality (AR) under the supervision of Professor Jivko Sinapov.
- Utilized human-in-the-loop reinforcement learning techniques to identify the most effective AR visualizations for enhancing robot obstacle navigation training, while analyzing the impact of AR on user training strategies.
- Learned how to develop and deploy a Flask web interface for real-time robot control, utilizing Docker for containerization and integrating with ROS to enhance user interaction and accessibility.

Projects

Bill Splitting Automation Tool | *Individual Project*

- Built automated bill-splitting application using Python and OpenPyXL that generates dynamic Excel workbooks with interconnected formulas, reducing group payment settlement time from hours to minutes.
- Designed scalable spreadsheet architecture with conditional logic that automatically adjusts to any group size and calculates proportional cost distribution including taxes and tips.

Catch It, Recycle It | *Runner-Up, Environmental Track, JumboHack 2024*

- Led the development of an interactive recycling-themed game with a team of five at Tufts University, aimed at raising environmental awareness and promoting sustainability.
- Employed agile methodologies to effectively manage project tasks and enhance the development process.

Pizza-Sticks | *1st Place, Battlebot Competition, Tufts Robotics Club*

- Collaborated with a team of four to engineer and construct Pizza-Sticks, a competitive battlebot, using AutoCAD for design and 3D printing for component fabrication.
- Assembled and tested the robot, ensuring optimal performance through iterative design and troubleshooting.

Certifications

Machine Learning Foundations e-Certificate by Cornell University (2024) [[view](#)]

GIAC Foundational Cybersecurity Technologies (GFACT) Certification (2023) [[view](#)]

Python for Everybody Specialization by University of Michigan (2022) [[view](#)]